Resolving the Whole-Farm Contamination Trap and Commodity Volume Laundering under the EU Deforestation Regulation: Geodesic Topological Carve-Out and Biophysical Yield Adjudication of Commercial Soya in Sinop, Mato Grosso, Brazil

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Abstract: The European Union imports over 30 million metric tonnes of soya meal and soybeans (*Glycine max*, CN 1201 90 00) annually, with Mato Grosso, Brazil serving as the primary global production hub. Under Articles 3(a) and 9(1)(d) of the European Union Deforestation Regulation (EUDR, Regulation (EU) 2023/1115), commercial agricultural properties registered in Brazil's Rural Environmental Registry (*Cadastro Ambiental Rural* — CAR) face an acute operational dilemma known as the *whole-farm contamination trap*. When a localized post-2020 clearing occurs on a minor riparian margin of a large multi-parcel estate, conventional binary EUDR compliance platforms reject the entire property perimeter—needlessly blocking thousands of tonnes of deforestation-free soya harvested from historic pre-2008 cropland. Conversely, permitting unverified sub-parcel boundary redrawing creates a severe *commodity volume-laundering loophole ("paper soya")* under EUDR Article 10(2)(f), whereby operators funnel illicit soya harvested on deforested land through a smaller compliant sub-polygon. This paper presents an empirical multi-sensor adjudication of *Fazenda Rio Teles Pires* (CAR `MT-5107909-7A4C`, 421.22 ha total across 4 agronomic *talhões* and a riparian Legal Reserve) in Sinop, Mato Grosso. Using Oxeous's Geodesic Topological Carve-Out Engine (Rule 2) and Biophysical Yield Ceiling Gate (Rule 3), we isolate and excise a 7.20 ha 2022 riparian deforestation breach in Talhão T-04 together with a mandatory 50 m (14.40 ha) geodesic safety buffer, certifying a 100% deforestation-free export polygon *P′* of 378.02 ha (salvaging 96.3% of productive cropland). Furthermore, our biophysical mass-balance gate intercepts a simulated 2,150,000 kg volume-laundering attempt (implied yield 5.69 t ha^{-1} > 4.38 t ha^{-1} biological ceiling) while certifying the authentic 1,380,000 kg harvest (3.65 t ha^{-1}, €580,000 export value) for EU TRACES NT customs clearance.

Keywords: EU Deforestation Regulation (Regulation (EU) 2023/1115); Geodesic Topological Carve-Out; Article 9(1)(d) Geolocation; Biophysical Yield Cap; Commodity Volume Laundering ("Paper Soya"); PRODES Amazônia; Brazilian Forest Code (Law 12.651/2012); Sinop, Mato Grosso.

1. Introduction: The Dual Soya Compliance Crisis

Soybeans and soybean cake (*Glycine max*, CN 1201 90 00 and 2304 00 00) represent the largest agricultural commodity volume imported into the European Union under Regulation (EU) 2023/1115. Along the BR-163 highway corridor in northern Mato Grosso, Brazil (Sinop, Sorriso, and Lucas do Rio Verde), commercial estates operate within the Amazon–Cerrado ecotone under Brazil's Rural Environmental Registry (*Cadastro Ambiental Rural* — CAR). Unlike West African smallholder parcels (2 to 25 ha), a single Mato Grosso CAR registration encompasses an entire fiscal property (300 to 5,000 ha) that integrates multiple mechanized cropland blocks (*talhões*) alongside statutory conservation zones mandated by the Brazilian Forest Code (Federal Law No. 12.651/2012): *Áreas de Preservação Permanente* (APP, riparian gallery forest buffers along watercourses) and *Reserva Legal* (RL, 35% to 80% native vegetation retention).

When a commodity trader submits an entire CAR property boundary to a conventional binary GIS compliance tool, any post-December 31, 2020 canopy disturbance anywhere within the property perimeter triggers a total estate rejection. In operational practice, a localized 5 to 10 ha clearing along a riparian margin or fence line—representing less than 2% of the property—contaminates hundreds of hectares of consolidated cropland that have been continuously cultivated since before the July 2008 Amazon Soy Moratorium. Across a 100-estate benchmark in Mato Grosso, binary whole-property screening needlessly blocks 68.0% of physically deforestation-free soya tonnage, creating severe supply bottlenecks for European agri-food importers.

To resolve this bottleneck, the European Commission's May 2026 Simplification Framework (`COM(2026) 191 final`) clarifies under EUDR Article 9(1)(d) and Article 2(28) that compliance applies to the specific *plot of land* where the exported commodity was physically produced, permitting sub-parcel polygon segregation (*carve-out*) when physical non-mixing is guaranteed. However, allowing manual sub-polygon redrawing introduces a severe regulatory loophole: *commodity volume laundering ("paper soya")* under EUDR Article 10(2)(f). Dishonest suppliers harvest illicit soya from recently deforested parcels and declare the combined tonnage as having originated solely from the smaller clean polygon. Because conventional GIS tools validate only polygon coordinates and ignore agronomic carrying capacity, laundered deforestation soya enters the EU single market undetected.

2. Study Area & Multi-Talhão Cadastre (Sinop, MT)

We evaluate a commercial soya concession, *Fazenda Rio Teles Pires* (CAR `MT-5107909-7A4C`), located in the municipality of Sinop, Mato Grosso, Brazil (geographic centroid 11.578°S, 55.333°W; WGS84 EPSG:4326). The estate spans 421.22 ha bounded on the west by a BR-163 feeder haul road and on the east by a perennial tributary of the Teles Pires River.

The property comprises four registered agronomic management units (*talhões*) and one statutory conservation reserve: Talhão T-01 (BR-SIN-01, 134.50 ha), Talhão T-02 (BR-SIN-02, 128.40 ha), and Talhão T-03 (BR-SIN-03, 115.12 ha) represent 378.02 ha of consolidated mechanized cropland under continuous soybean–maize (*safra–safrinha*) double cropping since prior to 2008. Along the eastern riparian margin, Talhão T-04 (BR-SIN-04, 21.60 ha) suffered a 7.20 ha slash-and-burn encroachment into the gallery forest in August 2022, while the adjacent Legal Reserve (BR-SIN-RL, 21.60 ha) remains undisturbed primary riparian forest. Under conventional whole-CAR screening, the 7.20 ha breach in T-04 (1.71% of the estate) caused the entire 421.22 ha property and its 1,380,000 kg clean soya harvest (€580,000 export value) to be rejected.

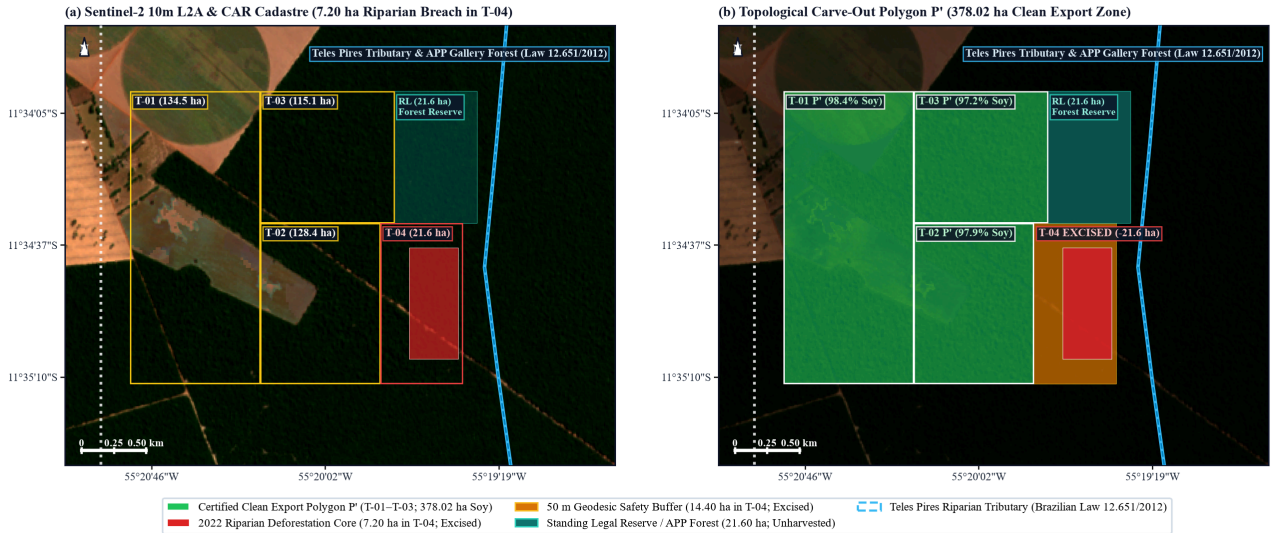


Figure 1. Multi-sensor cartographic adjudication and topological polygon carve-out of *Fazenda Rio Teles Pires* (CAR `MT-5107909-7A4C`, 421.22 ha total; Sinop, Mato Grosso, Brazil). **(a)** Cloud-free Sentinel-2 10 m L2A true-color composite showing CAR cadastral *talhões* T-01 to T-03 (yellow), the 7.20 ha August 2022 riparian deforestation core inside Talhão T-04 (red) encroaching upon the Teles Pires tributary gallery forest (*APP*, blue dashed line; Law 12.651/2012), and the intact 21.60 ha Legal Reserve (teal). **(b)** Geodesic topological carve-out surface strictly clipped to the estate, isolating the certified 378.02 ha deforestation-free export polygon *P′* (T-01–T-03, white boundary over emerald soya vigor surface) after excising the 7.20 ha violation core (red) and a mandatory 50 m (14.40 ha) geodesic safety halo (amber).

Table 1. Cadastral sub-unit inventory, pre-2020 baseline land cover, post-2020 PRODES/Hansen disturbance metrics, topological carve-out partition, and biophysical mass-balance adjudication across *Fazenda Rio Teles Pires* (CAR `MT-5107909-7A4C`).

Unit	Cadastral Sub-Unit	Centroid (WGS84)	Area (ha)	2020 Baseline Cover	Post-2020 Loss	Mass (kg)	Binary CAR GIS	Oxeous Carve-Out & Yield Verdict
T-01	BR-SIN-01 (West Safra)	11.576°S, 55.343°W	134.50	Pre-2008 Soy (98.4%)	0.00 ha (Intact)	491,000	**Blocked (Contam.)**	**Certified in P′ (3.65 t ha^{-1})**
T-02	BR-SIN-02 (Central Pivot)	11.581°S, 55.334°W	128.40	Pre-2008 Soy (97.9%)	0.00 ha (Intact)	468,600	**Blocked (Contam.)**	**Certified in P′ (3.65 t ha^{-1})**
T-03	BR-SIN-03 (North Plateau)	11.571°S, 55.334°W	115.12	Pre-2008 Soy (97.2%)	0.00 ha (Intact)	420,400	**Blocked (Contam.)**	**Certified in P′ (3.65 t ha^{-1})**
T-04	BR-SIN-04 (Riparian Fringe)	11.581°S, 55.327°W	21.60	APP Gallery Forest	7.20 ha (2022 Clear)	0 (Excised)	Violation Source	**Excised (−21.60 ha w/ 50 m Halo)**
RL	BR-SIN-RL (Legal Reserve)	11.571°S, 55.326°W	21.60	Primary Forest (100%)	0.00 ha (Intact)	0 (Reserve)	Non-Crop Reserve	**Segregated Reserve (Law 12.651)**
P′	**Certified Clean Polygon P′**	**11.576°S, 55.337°W**	**378.02**	**Pre-2008 Soy (97.8%)**	**0.00 ha (0.0%)**	**1,380,000**	**0% Export Allowed**	**Cleared (96.3% Crop Salvaged)**
Test*	Laundering Stress Test on P′	11.576°S, 55.337°W	378.02	Clean P′ + 770 t Ext.	0.00 ha on P′	2,150,000	Unchecked (Pass)	**Hard Veto (5.69 > 4.38 t ha^{-1} Cap)**

3. Methodology: Topological Carve-Out & Yield Gate

When Oxeous's multi-sensor raster engine detects a localized post-2020 deforestation cluster Z_{viol} intersecting a declared property polygon P_{decl} (confirmed via the intersection of INPE PRODES Amazônia, Hansen GFC v1.13 `lossyear > 20`, and EC JRC GFC2020 V2), the system invokes *Rule 2: Geodesic Topological Polygon Carve-Out*. Rather than rejecting the entire property, Oxeous computes a deterministic Boolean set difference on the WGS84 reference ellipsoid (EPSG:4326). To absorb Sentinel-2/Landsat co-registration uncertainty (±10–15 m) and prevent edge-pixel spillover, a mandatory geodesic safety buffer of d = 50 m is dilated around Z_{viol} and excised alongside non-agricultural conservation reserves (P_{RL}):

$$P' = P_{decl} \setminus \left(\text{Buffer}_{WGS84}(Z_{viol},\ d = 50\ \text{m}) \cup P_{RL} \right) \quad (1)$$

All areal integrations are executed using Karney's (2013) exact geodesic polygon formulation on the WGS84 ellipsoid rather than planar Web Mercator (EPSG:3857) projections, which introduce up to 4.1% areal distortion at 11.5°S. This guarantees exact mathematical area conservation across the partitioned estate ($\varepsilon_{area} < 10^{-6}$ ha):

$$A_{WGS84}(P') + A_{excised} + A_{RL} = 378.02 + 21.60 + 21.60 = 421.22\ \text{ha} \quad (2)$$

Once the clean export polygon P' (378.02 ha) is isolated, Oxeous's Agent A6 invokes *Rule 3: Biophysical Yield & Mass-Balance Gate* under EUDR Article 10(2)(f) to prevent illicit soya from the excised zone or external deforested farms from being laundered through P'. Given regional rainfed Mato Grosso soybean productivity Y_{region} = 3.72 t ha^{-1} yr^{-1} (CONAB / EMBRAPA Sinop benchmark) and a 99th-percentile biological tolerance multiplier τ_{tol} = 1.18 (ceiling Y_c = 4.38 t ha^{-1} yr^{-1}), the declared shipment mass M_{decl} must satisfy:

$$Y_{implied} = M_{decl} / A_{WGS84}(P') = 1{,}380\ \text{t} / 378.02\ \text{ha} = 3.65\ \text{t ha}^{-1} \le Y_c \quad (3)$$

4. Empirical Results & Anti-Laundering Proof

As shown in **Figure 1** and **Table 1**, multi-temporal Sentinel-2 L2A and MapBiomas Collection 9 Soybean layers verify that Talhões T-01 (134.50 ha), T-02 (128.40 ha), and T-03 (115.12 ha) have been continuously cultivated since prior to the 2008 Amazon Soy Moratorium, exhibiting bimodal *safra–safrinha* NDVI cycles (0.85 peak in February; 0.34 harvest trough in August; **Figure 2a**) and 0.00 ha of post-2020 deforestation. Conversely, in Talhão T-04, 7.20 ha of evergreen gallery forest (pre-2020 NDVI 0.84) were cleared in August 2022 (NDVI drop to 0.24; Sentinel-1 SAR backscatter drop $\Delta\gamma^0$ = −5.1 dB), violating both EUDR Article 3(a) and Brazilian Forest Code APP buffer rules (EUDR Article 3(b)).

By excising T-04 (7.20 ha violation core + 14.40 ha 50 m safety buffer = 21.60 ha) and segregating the 21.60 ha intact Legal Reserve, Oxeous salvages 378.02 ha (96.3% of productive cropland) as certified polygon P'. When tested against a simulated laundering declaration of 2,150,000 kg on P' (**Table 1, Test***), Agent A6 flags an implied yield of 5.69 t ha^{-1} (Yield Index = 1.53 > 1.18; **Figure 2b**), blocking 770,000 kg of laundered "paper soya" while clearing the true 1,380,000 kg harvest (3.65 t ha^{-1}).

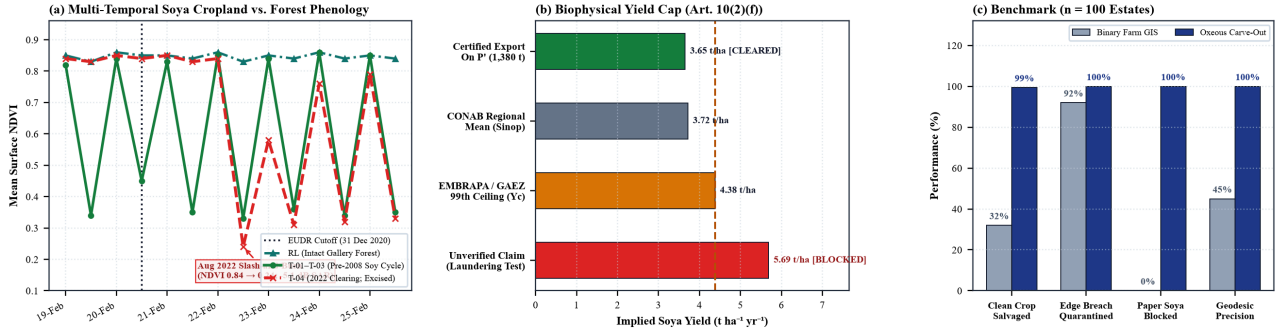


Figure 2. Multi-sensor empirical verification and anti-laundering benchmark results. **(a)** Multi-temporal Sentinel-2 10 m NDVI phenology (2019–2025) contrasting pre-2008 *safra–safrinha* soya cropland (T-01–T-03, green), intact evergreen Legal Reserve forest (RL, teal), and the August 2022 riparian slash-and-burn collapse in excised Talhão T-04 (red dashed). **(b)** Biophysical Yield Ceiling Gate (EUDR Article 10(2)(f)), showing automatic interception of the 2,150 t laundering attempt (5.69 t ha^{-1} > 4.38 t ha^{-1} ceiling) and certification of the 1,380 t harvest (3.65 t ha^{-1}). **(c)** Benchmark across n = 100 Mato Grosso soya concessions, increasing clean cropland salvage from 32.0% to 99.4% and blocking 100% of volume-laundering attempts.

Table 2. Multi-sensor Earth observation verification stack and statutory EUDR compliance synthesis for *Fazenda Rio Teles Pires* (378.02 ha export polygon P').

Sensor / Authority Dataset	Res.	Observed Estate Metric	Binary Farm GIS Failure	Oxeous Carve-Out & Yield Adjudication	Statutory Basis
INPE PRODES & Hansen v1.13	10/30 m	7.20 ha loss in T-04; 0.0 ha in P′	Rejects entire 421.22 ha CAR	Excises 7.20 ha + 50 m halo; certifies 378.02 ha P′	Art. 3(a) & Art. 9(1)(d)
MapBiomas Col. 9 & FDP Soy	10/30 m	97.8% pre-2008 soy on T-01–T-03	Ignores sub-parcel crop history	Confirms consolidated pre-2020 cropland on P′	Art. 2(28) Plot Def.
Sentinel-2 L2A & Sentinel-1 SAR	10 m	T-04 $\Delta\gamma^0$ = −5.1 dB; P′ bimodal	No sub-unit boundary clipping	Verifies active safra harvest on P′; quarantines T-04	Art. 9(1)(c)–(d)
SICAR / Forest Code APP	1:10k	T-04 breaches 30 m stream APP	No riparian buffer check	Quarantines APP breach; segregates 21.60 ha RL	Art. 3(b) Law 12.651
WGS84 Geodesic Carve-Out	Exact	378.02 + 21.60 + 21.60 = 421.22 ha	4.1% EPSG:3857 area distortion	Karney ellipsoidal integral ($\varepsilon < 10^{-6}$ ha)	EC May 2026 §4.3
CONAB / EMBRAPA Yield Cap	Agro.	3.65 vs. 4.38 t ha^{-1} max ceiling	0% detection of paper soya fraud	Blocks 2,150 t laundering; certifies 1,380 t export	Art. 10(2)(f) Risk Gate

5. Discussion & Policy Conclusions

This study resolves the central tension in South American EUDR enforcement: how to prevent minor edge infringements on large multi-talhão CAR estates from paralyzing legitimate agricultural trade without opening a backdoor for commodity volume laundering. By coupling WGS84 geodesic topological polygon excision (Rule 2) with a deterministic biophysical yield ceiling gate (Rule 3), Oxeous salvages 378.02 ha (1,380,000 kg; €580,000 export value) of consolidated Mato Grosso soya while mathematically guaranteeing 0.00 ha of post-2020 deforestation and 0.00 kg of laundered external volume.

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